



Department of Electrical and Electronic Engineering
Hardware / Software Report Submission

Semester: Summer 2026

Course Code: EEE101L

Course Title: Electrical Circuits I Laboratory

Section: 03

Experiment No: 4

Experiment Name: Lab Report_ Experiment 4

Group No: 01

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LAB REPORT: EXPERIMENT 4

TASK 1: VERIFICATION OF KCL AND KVL

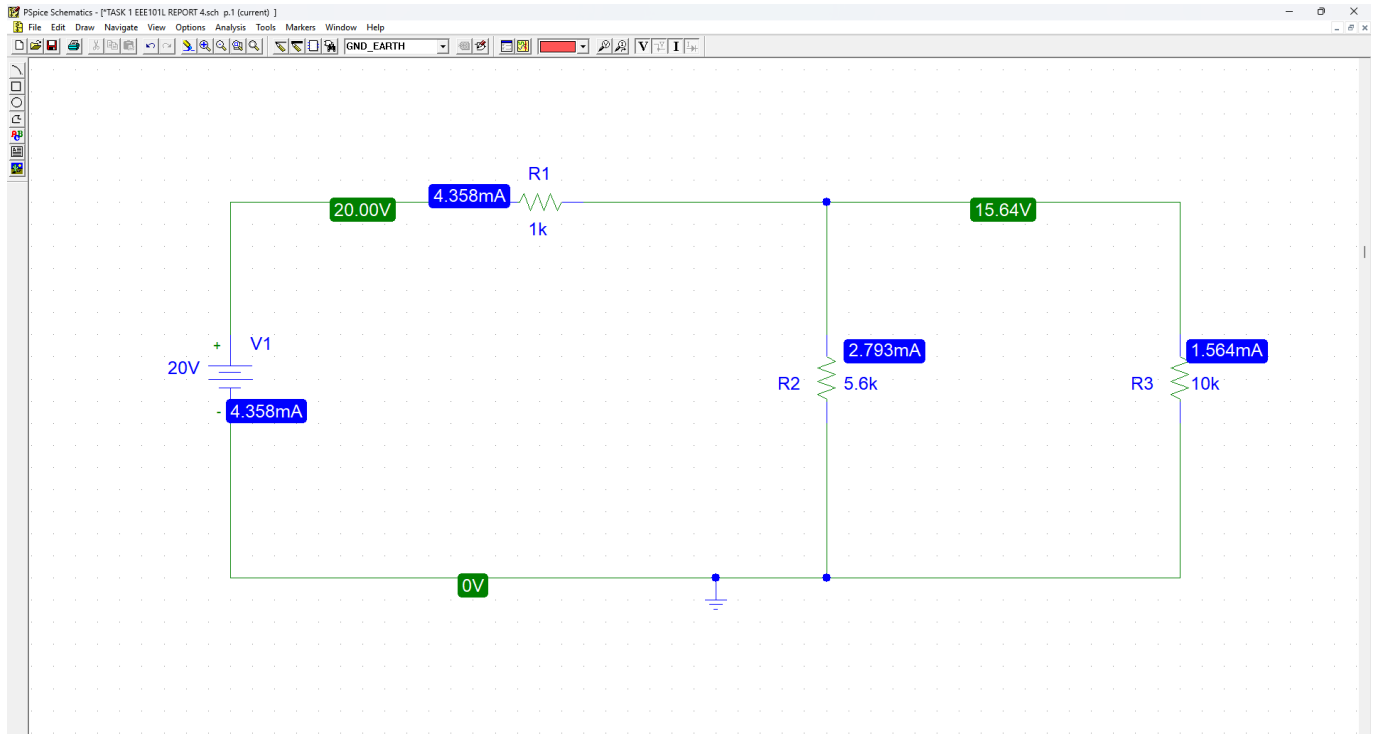


FIGURE 1: PSPICE SIMULATION FOR TASK 1

1.1 PSPICE MEASURED DATA TABLE

Parameter / Variable	Circuit Location	PSpice Measured Value
Vs	DC Source Voltage	20.00 V
Vnode	Node Voltage across R2 & R3	15.64 V
I1 (Input Current)	Current through R1 (from source)	4.358 mA
I2 (Branch Current 1)	Current through R2 (5.6k)	2.793 mA
I3 (Branch Current 2)	Current through R3 (10k)	1.564 mA

1.2 KCL AND KVL MATHEMATICAL PROOF TABLE

Law	Basic Relation	PSpice Mathematical Verification	Result
Kirchhoff's Current Law (KCL)	$I_{in} = I_{out1} + I_{out2}$	$4.358 \text{ mA} = 2.793 \text{ mA} + 1.564 \text{ mA} = 4.357 \text{ mA}$	VERIFIED
Kirchhoff's Voltage Law (KVL)	$V_s - V_{R1} - V_{R2} = 0$	$20.00 \text{ V} - 4.358 \text{ V} - 15.64 \text{ V} = 0.002 \text{ V} \approx 0 \text{ V}$	VERIFIED

TASK 2: VOLTAGE AND CURRENT ACROSS ALL RESISTORS

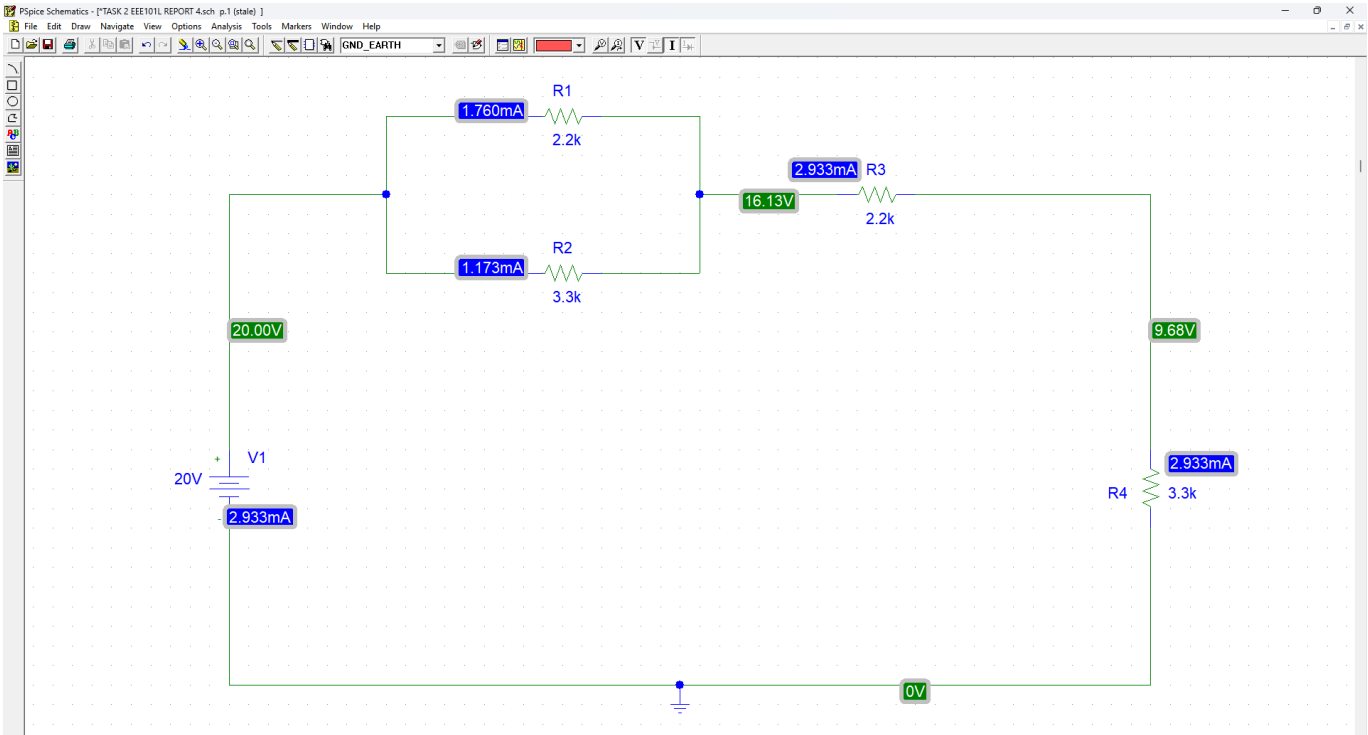


FIGURE 2: PSPICE SIMULATION FOR TASK 2

2.1 RESISTOR MEASUREMENT DATA TABLE

Resistor	Resistance Value	Voltage Across (V)	Current Through (mA)
R1	2.2 kΩ	3.87 V (20.00V - 16.13V)	1.760 mA
R2	3.3 kΩ	3.87 V (20.00V - 16.13V)	1.173 mA
R3	2.2 kΩ	6.45 V (16.13V - 9.68V)	2.933 mA
R4	3.3 kΩ	9.68 V (9.68V - 0.00V)	2.933 mA

2.2 CIRCUIT MATHEMATICAL CHECKS TABLE

Circuit Check	Basic Relation	PSpice Mathematical Verification	Result
Parallel Branch Current	$I_{\text{parallel}} = I_1 + I_2$	$1.760 \text{ mA} + 1.173 \text{ mA} = 2.933 \text{ mA} = I_3 = I_4$	VERIFIED
Total Voltage Balance	$V_{\text{total}} = V_{R1,2} + V_{R3} + V_{R4}$	$3.87 \text{ V} + 6.45 \text{ V} + 9.68 \text{ V} = 20.00 \text{ V} = V_s$	VERIFIED

TASK 3: SUPERPOSITION THEOREM VERIFICATION FOR V4

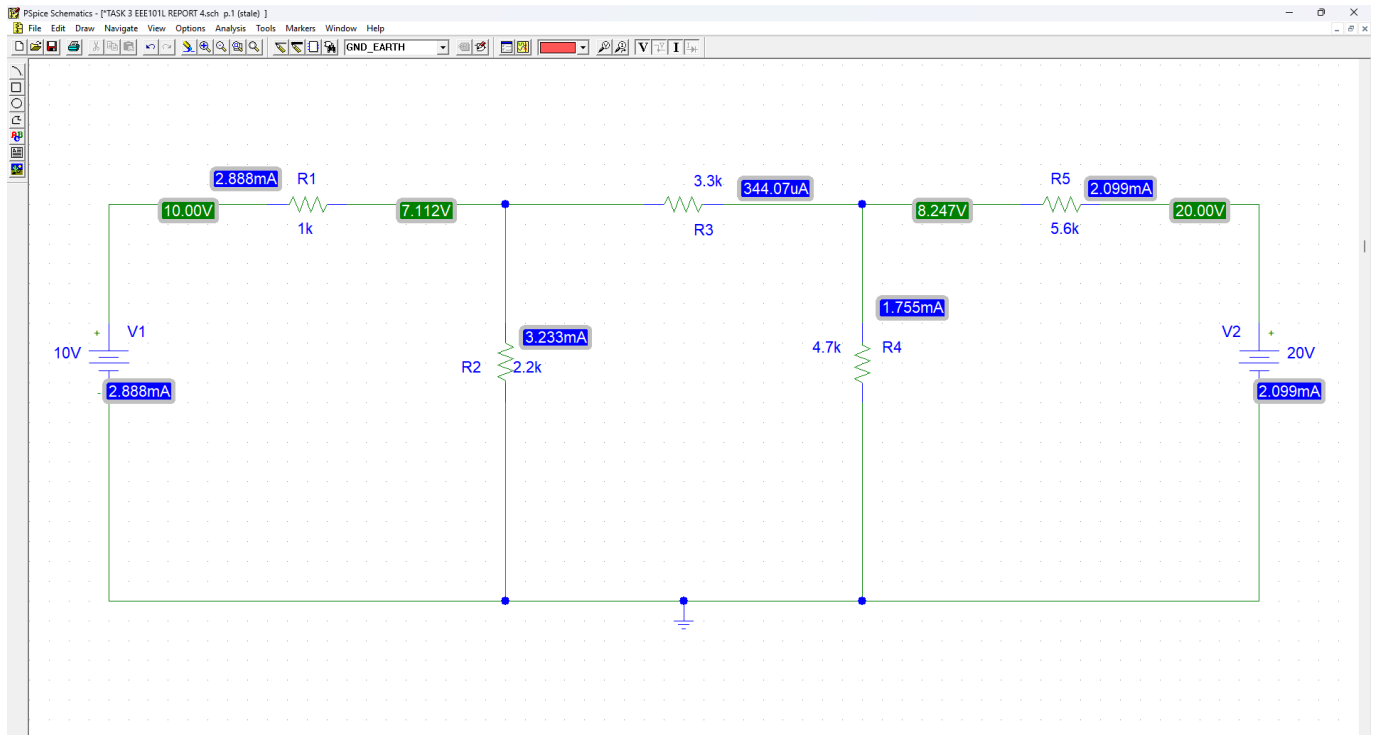


FIGURE 3.1: FULL CIRCUIT (BOTH SOURCES ACTIVE: V1 = 10V, V2 = 20V)

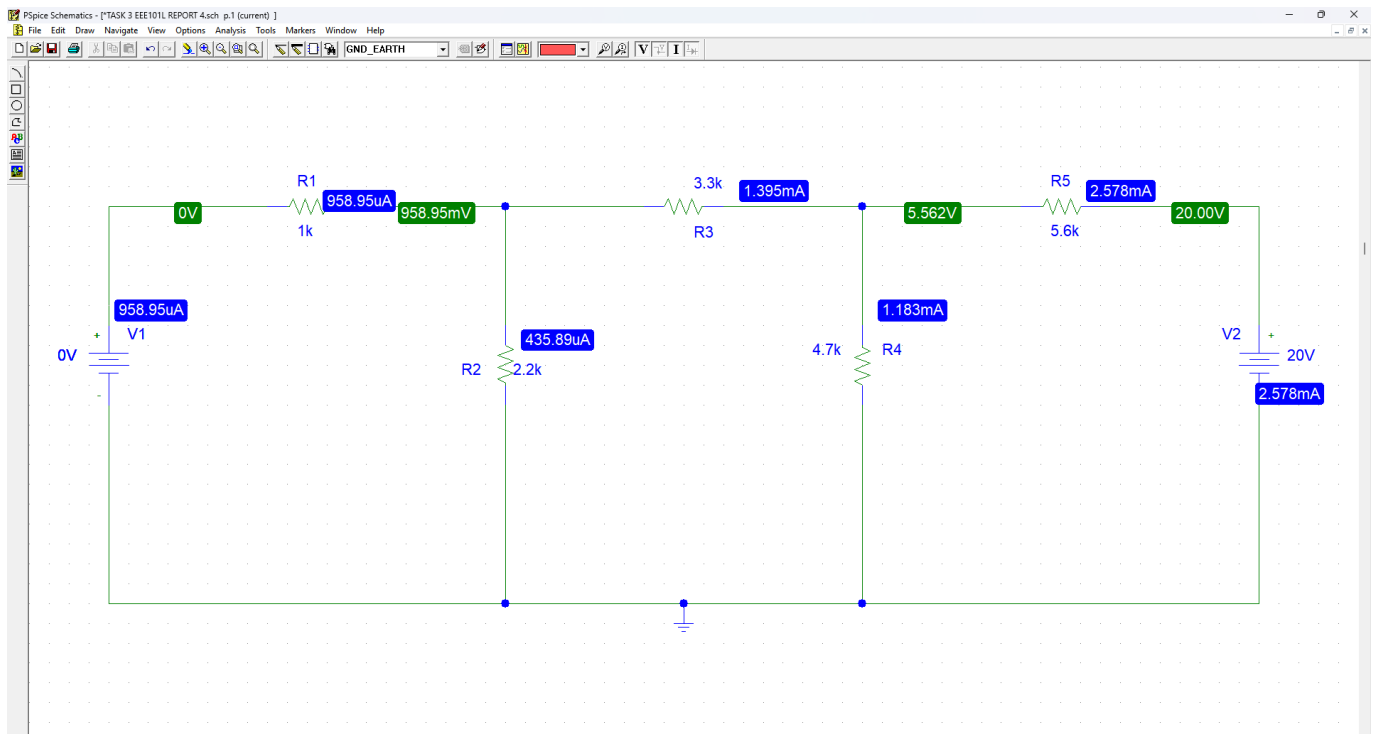


FIGURE 3.2: CASE 1 (10V SOURCE OFF / SHORTED, 20V ACTIVE)

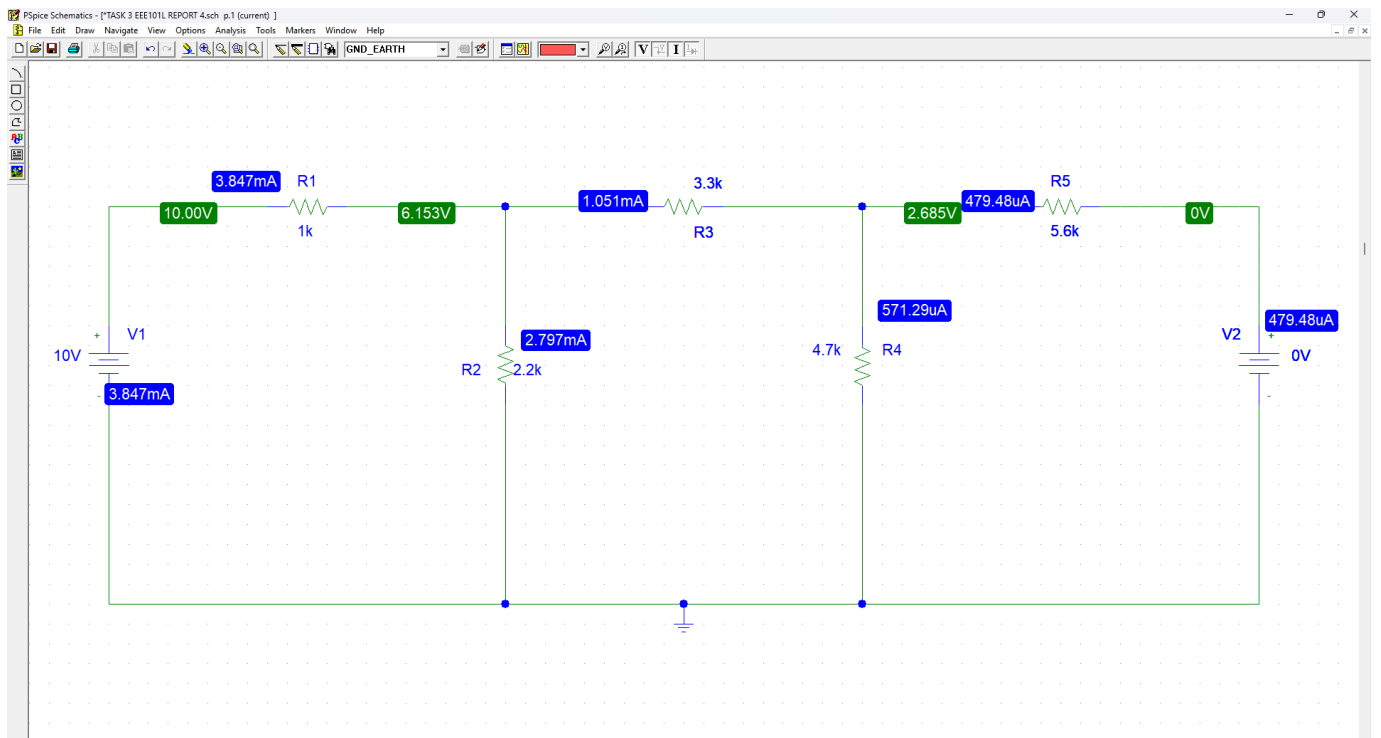


FIGURE 3.3: CASE 2 (20V SOURCE OFF / SHORTED, 10V ACTIVE)

3.1 SUPERPOSITION MEASURED DATA TABLE

Circuit State / Case	Source V1 (10V)	Source V2 (20V)	Measured V4 across R4 (4.7kΩ)
Case 1 (10V Source OFF)	0 V (Short Circuit)	20 V Active	$V4' = 5.562 \text{ V}$
Case 2 (20V Source OFF)	10 V Active	0 V (Short Circuit)	$V4'' = 2.685 \text{ V}$
Full Circuit (Both Active)	10 V Active	20 V Active	$V4 \text{ (Direct)} = 8.247 \text{ V}$

3.2 SUPERPOSITION MATHEMATICAL PROOF TABLE

Theorem Relation	Superposition Sum Formula	PSpice Mathematical Verification	Result
Superposition Principle	$V4 \text{ (Total)} = V4' + V4''$	$5.562 \text{ V} + 2.685 \text{ V} = 8.247 \text{ V}$	VERIFIED
Full Circuit Measurement	$V4 \text{ (Direct Measured)}$	8.247 V	VERIFIED